Global Electric Vehicle Market Analysis

Performance, Pricing, Battery Technology and Sales

SmartBridge Skill Wallet Project: Data Analytics with Tableau

Stack: MySQL, SQL, Tableau, Python, Flask

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Project link (GitHub)	https://github.com/Srajalsahu/Project-on-Electric-Vehicle-Market-Performance-Insights-by-Srajal-Sahu.git
Tableau Dashboard	https://public.tableau.com/app/profile/srajal.sahu/viz/EVMarketAnalysisbySrajalSahu/RegionPrice?publish=yes
Tableau Story	EV Market Analysis by Srajal Sahu | Tableau Public

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1. Problem understanding

Electric vehicles are becoming a mainstream product, but manufacturers, policy makers and buyers still struggle to compare the market. Models vary widely in price, range, battery chemistry, charging speed and sales. Without a clear view of these factors it is hard to know which vehicle types sell best, what drives price, and which battery technologies last longest.

Objectives

- Measure the size and structure of the EV market by category, manufacturer, region and country.
- Understand how price relates to range, battery capacity and motor power.
- Compare battery technologies (NMC, LFP, NCA, Sodium-Ion, Solid-State) on durability and charging.
- Identify high-value vehicles and the environmental impact (CO_2 saved).
- Deliver the findings as an interactive Tableau dashboard and story embedded in a Flask web app.

Business questions

- Which categories and manufacturers lead in annual sales?
- How strongly do battery capacity and range drive price?
- Which battery type has the best cycle life and lowest degradation?
- How do market regions and manufacturing countries compare?
- Which vehicles give the most range per dollar?
- How much CO_2 do EVs save, by category?

2. Data collection and dataset description

The dataset **ev_market_test.csv** was provided by Skill Wallet. It has **100,000 rows and 37 columns**; each row is one EV model or variant (vehicle_id is unique and there are no duplicate records). Model years run from 2018 to 2026 with about 11,000 models per year.

Group	Columns
Identity	vehicle_id, model_name, manufacturer (200), model_year, manufacturing_country (10), market_region (7), category (15)
Battery	battery_capacity_kwh, battery_type (5), battery_cycle_life, battery_degradation_pct_5yr
Performance	range_km, efficiency_wh_per_km, motor_power_kw, torque_nm, num_motors, drivetrain, top_speed_kmph, acceleration_0_100_sec
Charging	max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr, num_charging_ports
Body and features	num_seats, weight_kg, cargo_capacity_liters, infotainment_screen_inch, autonomous_level, safety_rating_stars
Commercial	price_usd, insurance_cost_usd_yearly, warranty_years, resale_value_pct_3yr, annual_sales_units, customer_rating
Impact and use	co2_saved_tons_per_year, recommended_use_case (7)

Data quality findings

Column	Missing values
resale_value_pct_3yr	3,057
battery_cycle_life	2,028
customer_rating	1,931
infotainment_screen_inch	1,506
annual_sales_units	989

- All other columns are complete. Outliers exist by design (buses and trucks have very high efficiency_wh_per_km, seats up to 65, price up to about $410k), so charts are segmented by category.
- The data is synthetic: models per year, regional totals and average range by battery type are almost uniform. Visuals focus on fields with real variation (category, price, battery durability).

3. Storing data in the database

MySQL was used as the database. Create the database, then load the CSV with pandas and SQLAlchemy (script `scripts/load_to_db.py`).

```
CREATE DATABASE IF NOT EXISTS ev_db;

# scripts/load_to_db.py (core)
df = pd.read_csv("data/raw/ev_market_test.csv")
df.to_sql("ev_market", engine, if_exists="replace", index=False, chunksize=5000)
print(f"Loaded {len(df):,} rows into ev_market")
```

Add screenshot here: MySQL Workbench showing the ev_market table and the row count (100,000).

4. SQL operations

The queries below are saved in `sql/queries.sql`. They cover aggregation, grouping, ranking, calculated columns, a null audit and a view for Tableau.

```
-- Top 10 manufacturers by sales
SELECT manufacturer, SUM(annual_sales_units) AS total_sales
FROM ev_market GROUP BY manufacturer ORDER BY total_sales DESC LIMIT 10;

-- Battery technology comparison
SELECT battery_type, ROUND(AVG(range_km),1) AS avg_range_km,
       ROUND(AVG(charging_dc_fast_time_min),1) AS avg_fast_charge_min,
       ROUND(AVG(battery_cycle_life),0) AS avg_cycle_life,
       ROUND(AVG(battery_degradation_pct_5yr),2) AS avg_degradation_pct
FROM ev_market GROUP BY battery_type ORDER BY avg_cycle_life DESC;

-- Best value vehicles
SELECT model_name, manufacturer, category, price_usd, range_km,
       ROUND(range_km / price_usd * 1000, 2) AS km_per_1000usd
FROM ev_market WHERE price_usd > 0 ORDER BY km_per_1000usd DESC LIMIT 20;

-- Clean view used by Tableau
CREATE OR REPLACE VIEW vw_ev_clean AS
SELECT *, ROUND(range_km / price_usd * 1000, 2) AS range_per_1000usd,
  CASE WHEN price_usd < 20000 THEN 'Budget' WHEN price_usd < 60000 THEN 'Mid-range'
       WHEN price_usd < 120000 THEN 'Premium' ELSE 'Luxury' END AS price_segment,
  price_usd * annual_sales_units AS est_revenue_usd
FROM ev_market WHERE price_usd > 0 AND range_km > 0;
```

Other queries in the file: sales by category, region summary, year trend, manufacturing country summary, use-case summary and the missing-value audit. *Add screenshots of at least three query outputs here.*

5. Connecting the database to Tableau

- Install the MySQL ODBC/JDBC driver for Tableau Desktop.
- Connect > To a Server > MySQL; enter server, port 3306, username, password.
- Select database **ev_db** and drag the view **vw_ev_clean** to the canvas.
- Use a **Live** connection for testing and an **Extract** for the final workbook.

Add screenshot here: Tableau data source page showing the connected view and 100,000 rows.

6. Data preparation

- **Nulls:** left blank for averages (Tableau ignores nulls). Missing values are not imputed because they are under 3.1% per column.
- **Filtering:** the view removes rows with non-positive price or range (none found in this dataset).
- **Data types:** model_year kept as a discrete dimension; geographic role set on manufacturing_country.
- **High cardinality:** 200 manufacturers are handled with a Top N parameter.
- **Derived fields:** Price Segment, Range Class, Fast Charge Rating, Range per $1000 and Revenue (Est.).
- **Outliers:** buses, trucks and scooters are compared within their category rather than removed.

Price segment distribution after preparation

Segment	Rule	Models
Budget	price < $20,000	26,426
Mid-range	$20,000 to $59,999	38,250
Premium	$60,000 to $119,999	20,902
Luxury	$120,000 and above	14,422

7. Visualizations and calculated fields

The workbook contains 12 unique visualizations, 6 calculated fields and 6 filters.

#	Visualization	Type	Question answered
1	Sales by category	Bar	Which vehicle types sell most?
2	Sales by model year	Line	Is the market growing?
3	Top N manufacturers	Bar + parameter	Who leads?
4	Manufacturing country	Filled map	Where are models built?
5	Region vs average price	Bar	How do regions compare?
6	Battery type share	Donut	Which chemistry dominates?
7	Cycle life and degradation	Bar / dual axis	Which battery lasts longest?
8	Price vs range	Scatter	Does more money buy more range?
9	Category x drivetrain	Heat map	Where do FWD/RWD/AWD concentrate?
10	Recommended use case	Treemap	What are EVs used for?
11	CO2 saved	Stacked bar	Where is the climate impact?
12	Customer rating	Histogram	How satisfied are owners?

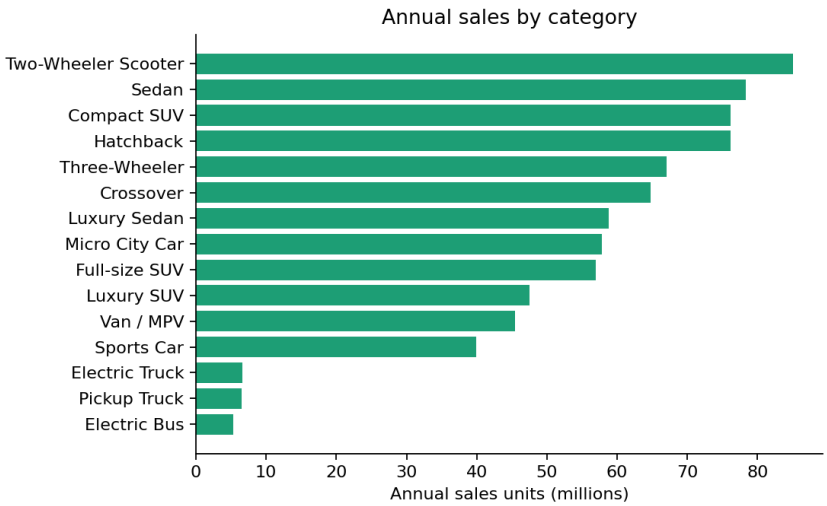
Calculated fields

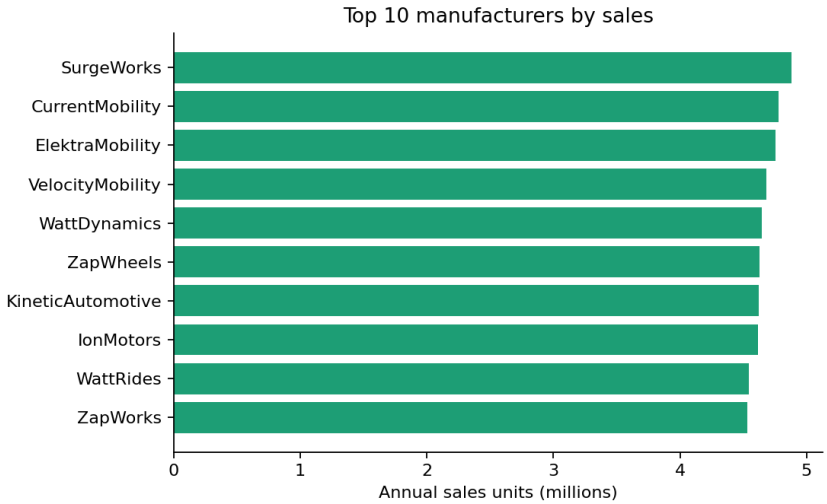
```
Range per $1000     = [range_km] / [price_usd] * 1000
Price Segment       = IF [price_usd] < 20000 THEN "Budget" ELSEIF [price_usd] < 60000 THEN "Mid-range"
                      ELSEIF [price_usd] < 120000 THEN "Premium" ELSE "Luxury" END
Range Class         = IF [range_km] >= 450 THEN "Long" ELSEIF [range_km] >= 250 THEN "Medium" ELSE "Short" END
Fast Charge Rating = IF [charging_dc_fast_time_min] <= 15 THEN "Ultra-fast"
                      ELSEIF [charging_dc_fast_time_min] <= 25 THEN "Fast" ELSE "Standard" END
Revenue (Est.)      = [price_usd] * [annual_sales_units]
Sales Share %       = SUM([annual_sales_units]) / TOTAL(SUM([annual_sales_units]))
```

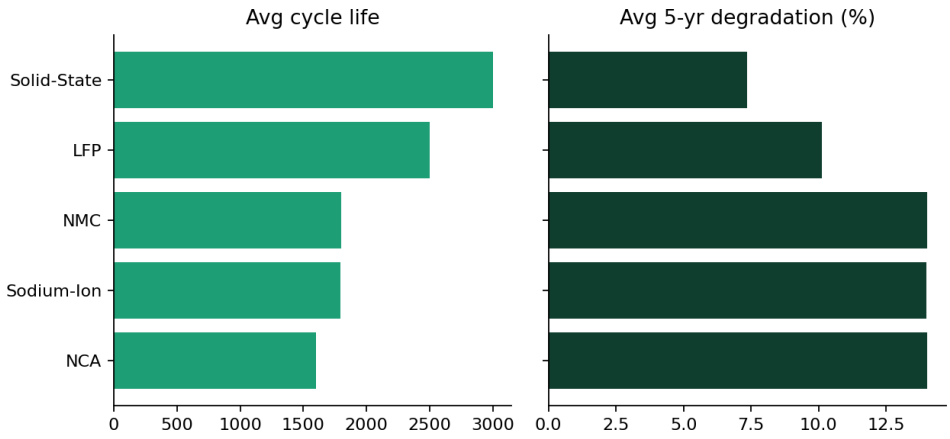
Filters: model_year, market_region, category, battery_type, Price Segment, Top N manufacturers (applied to all worksheets using the data source).

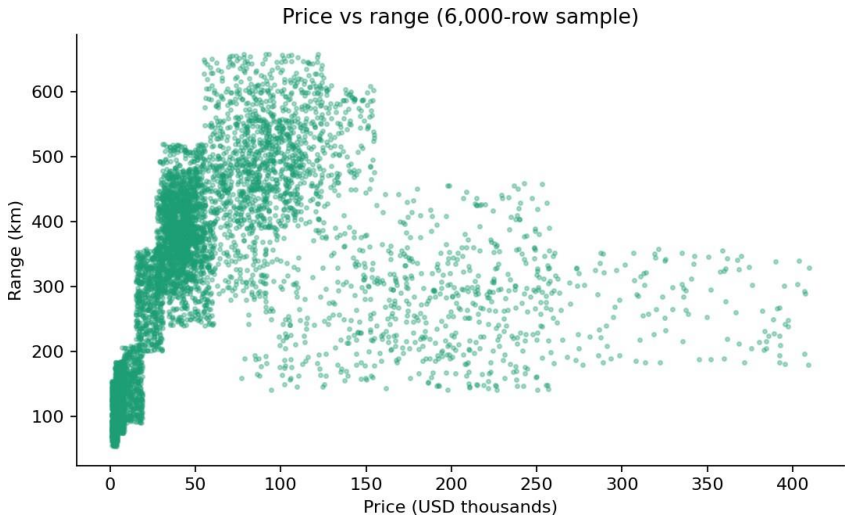
Sample visual outputs (reproduced from the dataset)

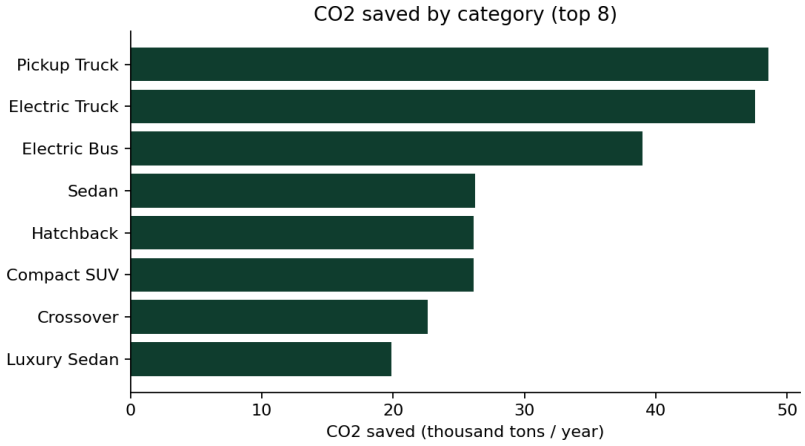
These charts were generated from the same data to show the expected patterns; rebuild them in Tableau and replace them with your own screenshots.






Avg cycle life
Avg 5-yr degradation (%)
Solid-State
LFP
NMC
Sodium-Ion
NCA
0
500
1000
1500
2000
2500
3000
0.0
2.5
5.0
7.5
10.0
12.5


Price vs range (6,000-row sample)
Range (km)
100
200
300
400
500
600
0
50
100
150
200
250
300
350
400
Price (USD thousands)


CO2 saved by category (top 8)
Pickup Truck
Electric Truck
Electric Bus
Sedan
Hatchback
Compact SUV
Crossover
Luxury Sedan
0
10
20
30
40
50
CO2 saved (thousand tons / year)

8. Dashboards

- **Market Overview:** KPI cards (total models, total sales, average price, average range) with sales by category, sales trend, top manufacturers and region vs price.
- **Technology:** battery share, cycle life vs degradation, price vs range scatter and the category x drivetrain heat map.
- **Geography and Impact:** manufacturing country map, use-case treemap, CO_2 saved and rating histogram.

Responsive design

- Dashboard size set to **Automatic** so it fits the browser window.
- Phone and Tablet layouts created with Device Preview.
- Layout containers (horizontal/vertical) keep charts aligned; filters sit in a collapsible side panel.
- Dashboard actions: Use as Filter on category and region charts; Highlight on battery type.
- One green palette with consistent fonts and clear titles for readability.

Add screenshots: desktop, tablet and phone layouts of each dashboard.

9. Story (6 scenes)

Scene	Title	Message
1	Market snapshot	100,000 models, average price about $65,010, average range about 324 km.
2	Who sells the most	Two-wheeler scooters, sedans and compact SUVs lead; trucks, pickups and buses sell far fewer units.
3	Price vs range	Battery capacity drives price much more than range does.
4	Battery technology	Solid-State and LFP last longest with the lowest degradation.
5	Regions and countries	Regional totals are balanced; no single region or builder dominates.
6	Recommendations	Focus on high-volume categories and durable battery chemistries.

10. Performance testing

Parameter	Value
Rows rendered from DB	100,000
Columns	37 (plus derived fields)
Data filters used	6
Calculated fields	6 (plus Top N rank field)
Visualizations	12
Dashboards / story scenes	3 / 6
Live connection load time	_____s (measure)
Extract load time	_____s (measure)

Measure load time with Help > Settings and Performance > Start Performance Recording, open each dashboard, then stop and screenshot the summary. Tips used to keep the dashboard fast: use an extract, hide unused fields, apply context filters, limit marks on scatter plots, and use the Top N parameter.

11. Web integration with Flask

Publish the workbook to Tableau Public and copy the view URLs into the `.env` file. The Flask app embeds them with the Tableau Embedding API.

```
# app.py (core)
@app.route("/dashboard")
def dashboard():
    return render_template("dashboard.html", viz_url=DASHBOARD_URL)

# templates/embed.html
<script type="module" src="https://public.tableau.com/javascripts/api/tableau.embedding.3.latest.min.js"></script>
<tableau-viz src="{{ viz_url }}" toolbar="bottom" hide-tabs></tableau-viz>
```

Routes: / (home), /dashboard, /story, /about. The CSS is responsive and supports light and dark mode. Run with `python app.py` and open http://127.0.0.1:5000.

Add screenshots: home page, dashboard page and story page running in the browser.

12. Key insights

- **Scooters lead volume:** Two-wheeler scooters sell about 85.0M units, followed by sedans (78.3M) and compact SUVs (76.1M). Electric trucks, pickups and buses sell only about 5 to 7M each.
- **Price follows battery size:** price correlates 0.84 with battery capacity but only 0.28 with range. Electric buses (average about $274k) and trucks (about $169k) are the most expensive categories.
- **Battery durability:** Solid-State averages about 2,999 charge cycles and 7.4% degradation over 5 years; LFP follows with about 2,498 cycles and 10.1%. NMC, NCA and Sodium-Ion are around 1,600 to 1,800 cycles and 14%.
- **No fast-charge or range gap:** every battery type averages about 21 minutes for DC fast charging and 320 to 325 km of range, so durability is the real differentiator.
- **Balanced market:** the top five manufacturers (SurgeWorks, CurrentMobility, ElektraMobility, VelocityMobility, WattDynamics) sell 4.6M to 4.9M units each, and regions and countries are evenly split.
- **Stable trend:** average price (about $65k) and range (about 324 km) are flat from 2018 to 2026, which suggests the data does not model technology improvement over time.

Limitation: because the data is synthetic, summed sales (about 773 million units) and estimated revenue are not real-world totals; treat them as relative measures.

13. Demo video plan, conclusion and future scope

Demo video outline (about 8 to 10 minutes)

- 0:00 Problem statement and objectives
- 1:00 Dataset overview
- 2:00 MySQL table load and SQL queries
- 3:30 Tableau connection and data preparation
- 4:30 Visualizations and calculated fields
- 6:00 Dashboards and responsive layouts
- 7:00 Story
- 8:00 Flask app running with embedded dashboard and story
- 9:00 Insights, performance results and conclusion

Conclusion

The project takes an EV market dataset from CSV to MySQL, analyses it with SQL, visualizes it in Tableau and publishes the dashboard and story in a Flask web app. The analysis shows which categories sell most, that battery size drives price, and that battery chemistry matters most for durability.

Future scope

- Add real-world data (registrations, charging stations) to replace synthetic fields.
- Add forecasting of sales by category.
- Add user login and a live database refresh to the web app.
- Deploy the Flask app to a cloud platform.